

TechArena 2025 Phase 2 Implementation Checklist (DAILY TRACKING)

Completed (Oct 25-31)

Phase 1 Foundation:

- ☒ Model refactoring and optimization
- ☒ Comprehensive 45-scenario validation
- ☒ Performance testing and constraint verification

Phase 2 Design:

- ☒ Mathematical formulation (`p2_model_ggdp.tex`)
- ☒ Degradation modeling approach
- ☒ aFRR energy market integration design

Data & Visualization Infrastructure:

- ☒ Data processing pipeline (`process_phase2_data.py`)
- ☒ View 1 visualizations (all 4 modules)
- ☒ McKinsey styling configuration (`viz_config.py`)
- ☒ Custom exceptions for error handling

Week 2 Checklist (Nov 1-9)

Day 1 (Nov 1): Degradation Model

- ☐ Extract parameters from Collath et al. (2023)
- ☐ Extract parameters from Xu et al. (2017)
- ☐ Create `battery_degradation.py`
- ☐ Implement `BatteryDegradationModel` class
- ☐ Test with dummy data
- ☐ Document assumptions

Day 2 (Nov 2): Phase 2 Model Part 1

- ☐ Create `model_phase2.py`

- ☐ Extend ImprovedBESSOptimizer → Phase2BESSOptimizer
- ☐ Add aFRR energy variables and constraints
- ☐ Add SOC segmentation variables
- ☐ Implement total power aggregation

Day 3 (Nov 3): Phase 2 Model Part 2

- ☐ Add SOS2 calendar aging variables
- ☐ Implement degradation cost in objective
- ☐ Update cross-market constraints
- ☐ **CRITICAL TEST:** DE, 0.5C, 2.0 must solve
- ☐ Verify revenue higher, net profit accounts for degradation

- [] **Next Review:** After implementation milestone (Nov 3 evening) or if major blocker occurs

Day 4-6 (Nov 4-6): Scenarios & Analysis

- ☐ Create run_phase2_scenarios.py
- ☐ Run 45 scenarios (background process)
- ☐ Handle failed scenarios
- ☐ Analyze results (best configs, degradation impact)
- ☐ Update investment analysis with NPV across all 5 countries

Day 7-9 (Nov 7-9): Documentation & Dashboard

- ☐ Create master results notebook
- ☐ Generate all visualizations
- ☐ Build interactive web dashboard (Plotly Dash, 3 tabs)
- ☐ Integrate existing McKinsey-style plot functions
- ☐ Write 15-page technical report
- ☐ Prepare submission CSVs
- ☐ Package code with documentation and dashboard
- ☐ Final validation & submission

You can do this. The math is done. The foundation is validated. The infrastructure exists. Now it's focused execution with clear deliverables.

Remember: 85% is a strong grade. Work smart with existing infrastructure. Execute efficiently. 🚀